Online PAC Labeling from a Linear Programming Perspective

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Recap: AI-assisted labeling

We have a pool of unlabeled items $X_1, \ldots, X_T$. Need to collect their true labels $Y_1, \ldots, Y_T$.

Two ways to get a label $\tilde{Y}_t$:

- **ML models** $\widehat{Y}_t^{(a)}$ — sometimes *wrong*, but *cheap*
- **Expert** Y_t — *correct*, but *expensive*.

Two Goals:

- Save cost by reducing the number of expert queries.
- Probably Approximately Correct (PAC) guarantee:

$$\frac{1}{T}\sum_{t=1}^{T} \ell(Y_t, \widetilde{Y}_t) \le \varepsilon \quad w.p. \ge 1 - \alpha.$$

Problem setup

- For each item X_t, decide how to label it: use the **expert** or one of k **cheap models**.
- Choose action a_t from an action set $\mathcal{A} = \{E, 1, \ldots, k\}$.
- For item t and action a, a labeling **loss** and a **cost**:

$$h_t(a) = \begin{cases} 0, & a = E \\ \ell(Y_t, \widehat{Y}_t^{(a)}), & a \in [k] \end{cases} \qquad c_t(a) = \begin{cases} 1, & a = E \\ 0, & a \in [k] \end{cases}$$

- Label the item with $\widetilde{Y}_t = Y_t$ if $a_t = E$, else $\widehat{Y}_t^{(a_t)}$.

Key quantities: *cost* we want to minimize, *error* we must control.

The cost-aware labeling LP

Let $x_{t,a} \in [0,1]$ be the probability of taking action a on item t.

$$\min_{x} \quad \underbrace{\frac{1}{T}\sum_{t}\sum_{a} c_t(a)\, x_{t,a}}_{\text{average cost}}$$

$$\text{s.t.} \quad \underbrace{\frac{1}{T}\sum_{t}\sum_{a} h_t(a)\, x_{t,a} \le \varepsilon,}_{\text{average error budget}} \qquad \sum_{a} x_{t,a} = 1, \quad x_{t,a} \ge 0.$$

- **One** global constraint: keep average error $\le \varepsilon$.
- Naturally handles *multiple* cheap sources and item-specific costs.
- If true losses $h_t(a)$ were known, this LP is the *optimal* policy.

The dual gives a single "price of error"

One Lagrange multiplier $\lambda \geq 0$ for the error constraint. The dual:

$$\max_{\lambda \geq 0} \; -\lambda\varepsilon + \frac{1}{T}\sum_{t=1}^{T} \min_{a \in \mathcal{A}} \left\{ c_t(a) + \lambda\, h_t(a) \right\}.$$

The fixed-price rule

Once λ is fixed, the decision **separates across items**:

$$a_t(\lambda) \in \arg\min_{a \in \mathcal{A}} \left\{ c_t(a) + \lambda\, h_t(a) \right\}.$$

λ is the exchange rate between cost and error.
Small $\lambda \Rightarrow$ trust AI | Large $\lambda \Rightarrow$ defer to expert.

What the price actually does: a threshold

With equal cheap-source costs, first **route** to the best cheap model

$$j_t^\star \in \arg\min_{j\in[k]} \widehat{h}_t(j), \qquad s_t = \min_j \widehat{h}_t(j) \quad \text{(the "uncertainty score")}.$$

Then λ only decides *accept vs. defer* via a threshold $1/\lambda$:

$$\boxed{s_t \geq 1/\lambda \;\Rightarrow\; \text{defer to expert}} \qquad s_t < 1/\lambda \;\Rightarrow\; \text{accept cheap label.}$$

- Recovers the original **PAC-labeling** threshold rule (single source: $\widehat{h}_t$ = uncertainty).
- Multi-source extension is "free": just replace the score by $\min_j \widehat{h}_t(j)$.
- Two regimes ahead: **offline** (fix λ once) and **online** (move λ_t over time).

Part I — Offline PAC Labeling

Certify one price on a calibration set, then label the whole pool.

Offline: certify a price from a small labeled sample

Idea. The cost $C_T(\lambda)$ is known from proxies; only the error $L_T(\lambda)$ needs true labels. Estimate it on a calibration set.

- Sample m items, get expert labels, form empirical loss $\widehat{L}_m(\lambda)$.
- Hoeffding upper confidence bound:

$$U_m(\lambda; \alpha) = \widehat{L}_m(\lambda) + \sqrt{\frac{\log(1/\alpha)}{2m}}.$$

- Pick the **cheapest certified price**:

$$\hat{\lambda} = \min\{\lambda \geq 0 : \; U_m(\lambda; \alpha) \leq \varepsilon\}.$$

Why no union bound over λ?

As λ grows, items only move *AI* $\rightarrow$ *expert*. So $L_T(\lambda)$ is **monotone** — a single nested family, certified for free.

Offline: the PAC guarantee

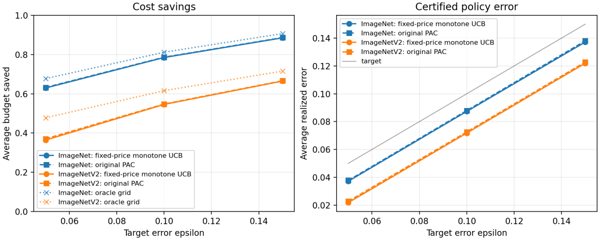
Theorem (offline monotone-threshold PAC)

With probability at least $1-\alpha$ over the calibration draw, the released labels satisfy

$$\frac{1}{T}\sum_{t=1}^{T}\ell(Y_t, \widetilde{Y}_t) \;\leq\; \varepsilon.$$

- Distribution-free, finite-sample. Falls back to expert-only if no finite price qualifies.
- **Corollary:** on a fresh deployment pool of size n, error $\leq \varepsilon + \sqrt{\log(1/\delta)/2n}$ w.p. $1-\alpha-\delta$.
- Proxies affect *cost*, not *validity*: a bad proxy just defers more often.

Offline experiments: LP rule matches original PAC



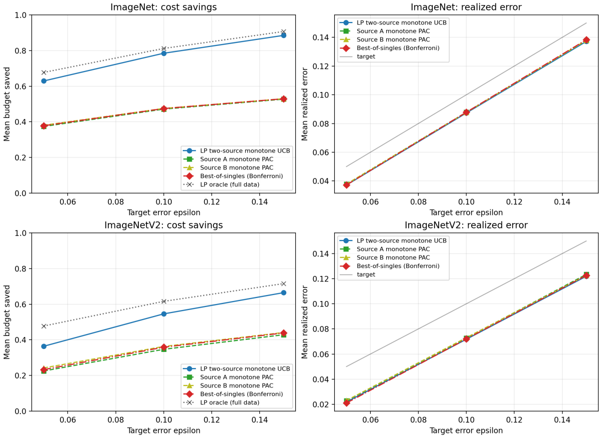
One cheap source, ImageNet / ImageNetV2.

One source ($\varepsilon = 0.10$):

Data	Method	Error	Saved
IN	LP	0.087	0.785
IN	PAC	0.088	0.786
INv2	LP	0.072	0.546
INv2	PAC	0.073	0.548

- Error stays under target ε (0 violations).
- LP fixed-price $\approx$ original PAC, as predicted.
- Saves $\sim$78% of expert budget on ImageNet.

Offline experiments: routing beats single sources



Two complementary cheap sources (by class parity).

Two sources ($\varepsilon = 0.10$, ImageNet):

Method	Error	Saved
LP routing	0.087	**0.785**
Best-of-singles	0.088	0.475

- Per-item routing to the lower-proxy source.
- Same error control, but **much** larger budget savings (0.78 vs 0.47).
- This is where the LP view earns its keep.

Part II — Online PAC Labeling via Audits

Items stream in; labels are hidden unless we audit.

Online: a moving threshold + randomized audits

Items arrive one at a time. AI losses are *unobserved* unless we pay the expert.

- Keep a moving price λ_t; route then threshold: $D_t = \mathbf{1}\{\widehat{h}_t^{\lambda} \geq 1/\lambda_t\}$ (defer) .
- If we use the AI label, **audit** it with probability $p_t \in [p_{\min}, 1]$:

$$A_t \sim \text{Bernoulli}(p_t).$$

If audited, query the expert, *correct* the label, and observe the AI's true error.

Inverse-probability feedback

$$\widehat{Z}_t = \frac{O_t}{q_t}(H_t - \varepsilon_0), \qquad \mathbb{E}[\widehat{Z}_t \mid \mathcal{G}_t] = H_t - \varepsilon_0 \ \ \text{(unbiased)}.$$

Audited rare errors are up-weighted by $1/p_t$ to stay unbiased.

Online: the price update

$$\lambda_{t+1} = \left[\lambda_t + \eta\,\widehat{Z}_t\right]_+.$$

- An audited **mistake** pushes λ_t *up* $\Rightarrow$ threshold $1/\lambda_t$ drops $\Rightarrow$ routing gets more conservative.
- A **deferral** or a clean audit pushes λ_t *down* $\Rightarrow$ trust the AI more.
- Internal target $\varepsilon_0 \le \varepsilon$ leaves slack for the finite-horizon bound.

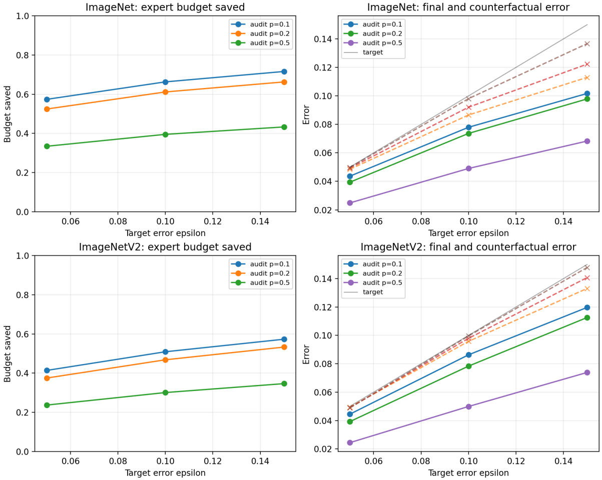
Theorem (audited online finite-horizon control)

With probability $\ge 1-\delta$ over audit randomness, for *any* data sequence,

$$\frac{1}{T}\sum_{t=1}^{T} G_t \;\le\; \varepsilon_0 + \frac{\Lambda_{\mathrm{on}}}{\eta T} + \frac{1}{p_{\min}}\sqrt{\frac{2\log(1/\delta)}{T}}.$$

Choosing $\eta \asymp T^{-1/2}$ gives final error $\le \varepsilon$, slack $O(T^{-1/2})$.

Online experiments: the validity–cost knob



Audited online sweep. Dashed = AI error before correction.

Online ($\varepsilon = 0.10$):

Data	p	Error	Saved
IN	0.1	0.078	0.663
IN	0.2	0.074	0.612
IN	0.5	0.049	0.395
INv2	0.2	0.078	0.468

- Smaller p: save more budget.
- Larger p: correct more, lower final error.
- Released-label error tracked below target under partial feedback.

Takeaways

One idea, two regimes

- LP dual $\Rightarrow$ a single *price of error* λ.
- **Offline:** certify $\hat{\lambda}$ on held-out labels.
- **Online:** move λ_t with audited inverse-probability feedback.

What we showed

- Proxies affect *cost*, not *validity*.
- LP rule matches PAC with one source; *routing wins* with two.
- Online stays valid under partial feedback; p trades cost vs. error.

Limitation & next step

Current data has only one real cheap source. The interesting case is many sources with *heterogeneous costs*, where a scalar threshold no longer suffices.

Thank you! **Questions?**